1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist identified that lipid/metabolic programs (HDL-C), stromal–adipocyte phenotypes (SARIFA), and mismatch-repair status (MSH6) co-occur as candidate biomarkers in MIBC; the core puzzle is why MSH6-linked immunogenicity (which should favor immune checkpoint responses) fails in some tumors that instead show adverse outcomes. 2) Tier 1 paths used: I built on MSH6 (DNA MMR → immunogenicity) and HDL-C (tumor lipid metabolism → immune modulation) because they form a direct mechanistic tension: MSH6 deficiency predicts higher neoantigen load and ICI sensitivity, while tumor-associated lipid handling (HDL-C-related) can create a metabolically immunosuppressive niche. I selected the imaging surrogate 'stromal-to-tumor ratio (TSR)' to capture the SARIFA/stroma–adipocyte context morphologically. 3) Tier 2 evidence supporting the claim: MSI status and AI-based biomarker paths support the link between MMR defects and ICI response; PTGDS and TGF-β paths support paracrine inflammatory/lipid signaling and stromal fibrosis that can reinforce metabolic immune evasion; lymph node metastasis provides a clinically relevant readout of aggressive biology. 4) Why therapy_response was chosen as the single endpoint: MSH6/MSI biology has the strongest direct mechanistic and clinical link to immune checkpoint responsiveness, and the key translational question is when predicted ICI sensitivity is negated by microenvironmental metabolic/stromal features. 5) Deliberate exclusions: I limited anchor variables to MSH6 and HDL-C to keep the hypothesis sharp; SARIFA (though core) and other supporting nodes (PTGDS, AI signatures, TGF-β, nodal status) were relegated to secondary_modifiers to preserve mechanistic focus and testability. I excluded broader prognostic endpoints (OS/PFI) to avoid conflating response mechanisms with downstream survival effects?

Working Hypothesis

Primary hypothesis: In tumors with high stromal-to-tumor ratio (TSR), HDL-C–driven adipocyte-like stromal metabolism creates an immunosuppressive microenvironment that blunts the enhanced immunogenicity conferred by MSH6 deficiency, resulting in reduced objective therapy_response to PD-1/PD-L1 blockade compared with MSH6-deficient tumors that have low TSR or low HDL-C pathway activity. The effect of MSH6 deficiency on ICI response is therefore moderated by TSR/HDL-C status, such that: (1) MSH6-deficient, low-TSR/low HDL-C tumors exhibit the strongest ICI responses; (2) MSH6-deficient, high-TSR/high HDL-C tumors show attenuated responses; (3) adding stromal/metabolic modulation could restore responsiveness in the high-TSR/MSH6-deficient subset. This interaction should be observable as a significant MSH6 × TSR/HDL-C interaction for therapy_response (ORR) and related clinical endpoints, after adjusting for MSI/TMB and other confounders. Secondary modifiers (SARIFA, MSI status, PTGDS, lymph node metastasis, AI signatures, TGF-β) may influence the magnitude of the interaction and can be explored to refine the predictive signature.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	MSH6, HDL-C
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6 (DNA MMR → immunogenicity) and HDL-C (tumor lipid metabolism → immune modulation) because they form a direct mechanistic tension: MSH6 deficiency predicts higher neoantigen load and ICI sensitivity, while tumor-associated lipid handling (HDL-C-related) can create a metabolically immunosuppressive niche.

HDL-C: HDL-C), stromal–adipocyte phenotypes (SARIFA), and mismatch-repair status (MSH6) co-occur as candidate biomarkers in MIBC; the core puzzle is why MSH6-linked immunogenicity (which should favor immune checkpoint responses) fails in some tumors that instead show adverse outcomes.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	SARIFA	Supporting modifier from knowledge graph
Secondary	MSI_status	Supporting modifier from knowledge graph
Secondary	PTGDS	Supporting modifier from knowledge graph
Secondary	lymph_node_metastasis	Supporting modifier from knowledge graph
Secondary	AI_based_biomarkers	Supporting modifier from knowledge graph
Secondary	TGF-β	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- SARIFA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSI_status: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.